

GOSAN SUWOLBONG, Korea, Republic of

WMO: 471850

Lat: 33.2939N

Lon: 126.1628E

Elev: 76

StdP: 100.42

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 43263

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.3	1.3	-6.7	2.2	3.4	-5.6	2.4	3.2	25.2	3.0	23.3	3.0	14.9	340	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.1	30.5	26.3	29.7	26.1	28.8	25.8	27.3	29.1	26.7	28.6	26.2	28.1	5.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.9	22.7	28.4	26.2	21.8	28.0	25.6	21.1	27.6	87.3	29.2	84.5	28.7	82.2	28.2	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.0	18.5	16.0	-1.1	32.1	1.6	1.1	-2.2	32.9	-3.2	33.5	-4.1	34.1	-5.3	34.9		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-2.7	27.7	1.1	0.9	-3.5	28.4	-4.2	28.9	-4.8	29.4	-5.6	30.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	6.0	6.6	9.4	13.5	17.2	20.8	25.1	26.6	23.2	18.7	13.6	8.4
	DBStd	7.42	2.98	3.16	2.93	2.53	2.02	1.68	2.24	1.80	1.94	2.47	3.19	3.12
	HDD10.0	358	130	103	47	3	0	0	0	0	0	6	69	
	HDD18.3	1663	384	329	278	146	47	2	0	0	24	145	307	
	CDD10.0	2478	5	7	27	108	222	325	468	516	397	271	113	20
	CDD18.3	741	0	0	0	1	11	77	210	257	147	37	2	0
	CDH23.3	4697	0	0	0	1	4	86	1536	2433	597	39	1	0
	CDH26.7	960	0	0	0	0	0	1	291	613	53	1	0	0
Wind	WSAvg	7.0	9.9	9.1	8.2	6.7	5.5	4.5	5.2	5.3	5.3	6.5	7.8	9.6
Precipitation	PrecAvg	1321	48	56	87	106	125	170	190	224	140	58	72	44
	PrecMax	2082	106	130	154	237	333	316	625	537	442	181	226	113
	PrecMin	776	9	9	12	30	54	40	10	60	13	3	2	8
	PrecStd	284	25	29	35	53	68	76	137	119	100	39	55	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.8	16.4	18.1	21.4	23.9	26.4	31.3	32.2	29.4	26.0	22.1	18.2
		MCWB	13.7	14.5	14.6	17.9	18.8	22.7	26.7	26.6	25.5	22.7	19.1	15.6
	2%	DB	13.8	14.9	16.3	19.5	22.5	25.2	30.1	31.0	28.1	24.4	20.8	16.0
		MCWB	11.4	13.1	14.0	16.3	18.5	22.1	26.5	26.4	24.7	20.9	18.1	12.7
	5%	DB	11.9	13.1	15.2	18.5	21.5	24.3	29.2	30.2	27.1	23.3	19.5	14.5
		MCWB	9.0	10.7	13.0	15.6	17.9	21.7	26.1	26.1	24.0	19.8	16.7	11.4
	10%	DB	10.4	11.5	13.9	17.4	20.5	23.5	28.3	29.4	26.2	22.1	18.2	13.0
		MCWB	7.8	9.1	11.5	14.7	17.4	21.3	25.7	25.9	23.3	18.7	15.5	10.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.7	15.2	15.9	19.0	21.2	24.9	28.4	28.3	26.7	24.0	20.5	16.7
		MCDB	15.4	16.0	17.0	20.0	22.0	25.6	30.0	30.2	28.1	25.1	21.6	17.6
	2%	WB	11.6	13.5	14.7	17.8	19.9	23.5	27.3	27.5	25.7	22.2	18.7	13.5
		MCDB	13.3	14.5	15.9	18.9	21.3	24.2	28.8	29.4	27.2	23.6	20.3	15.3
	5%	WB	9.5	11.2	13.4	16.5	19.1	22.7	26.6	26.9	25.0	20.7	17.2	11.7
		MCDB	11.4	12.6	14.8	17.7	20.5	23.5	28.2	29.0	26.5	22.4	19.0	14.1
	10%	WB	7.9	9.3	12.0	15.4	18.2	22.0	26.1	26.5	24.0	19.4	15.9	10.3
		MCDB	10.1	11.3	13.6	16.8	19.8	23.0	27.8	28.5	25.7	21.6	17.9	12.8
Mean Daily Temperature Range	5% DB	MDBR	3.6	4.3	4.9	5.0	4.7	4.0	3.5	4.1	4.1	4.2	4.4	3.9
		MCDBR	6.3	6.7	6.7	6.2	6.0	4.9	4.5	5.0	5.1	5.6	5.9	6.1
		MCWBR	5.8	5.9	5.8	4.8	4.3	3.4	2.3	2.6	3.3	4.5	5.1	5.7
	5% WB	MCDBR	6.0	6.4	6.4	5.1	4.9	3.5	3.8	4.2	4.3	4.8	5.5	5.9
		MCWBR	6.0	6.6	6.3	4.7	4.0	2.9	2.3	2.6	3.2	4.6	5.4	6.1
Clear-Sky Solar Irradiance	taub	0.465	0.504	0.553	0.561	0.568	0.598	0.582	0.536	0.492	0.466	0.464	0.460	
	taud	1.947	1.886	1.787	1.824	1.861	1.880	1.970	2.109	2.168	2.101	2.026	1.964	
	Ebn at Noon	707	727	729	748	748	723	732	760	777	758	706	684	
	Edh at Noon	150	176	209	210	205	200	182	157	142	140	137	140	
All-Sky Solar Radiation	RadAvg	1.98	2.85	4.09	4.86	5.20	4.58	5.03	5.20	4.32	3.75	2.56	1.85	
	RadStd	0.24	0.36	0.33	0.40	0.56	0.55	0.70	0.52	0.40	0.38	0.29	0.19	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page